H-Matrix BEM for Rough-Surface Normal Contact

Non-periodic solver · Love element integration · Polonsky–Keer PCG

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H-Matrix BEM Solver

The N^2 Wall

Surface contact requires **dense matrix–vector products**

$$u_i = \sum_{j=1}^{N} S_{ij}\, p_j, \qquad S_{ij} = \frac{1}{\pi E^*}\mathcal{L}\left(\mathrm{x}_i - \mathrm{x}_j, \tfrac{h}{2}\right)$$

Cost of dense BEM ($N = N_s^2$)

Operation	Time	Memory
Assembly	$O(N^2)$	$8N^2$ bytes
Matvec	$O(N^2)$	

Concrete: $N_s = 256$ ($N = 65\,536$) $\Rightarrow$ 34 GB just to store S

H-matrix target

Operation	Cost
Assembly	$O(N \log^2 N)$
Matvec	$O(N \log N)$
Memory	$\ll N^2$

Key idea: **far-field interactions are low-rank**

$$S[I, J] \approx UV^{\mathsf{T}}, \quad k \ll |I|, |J|$$

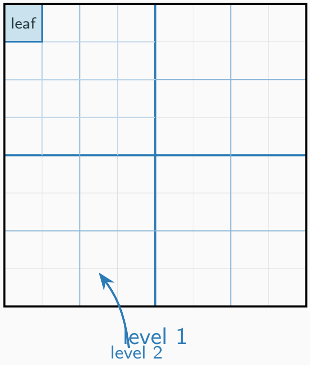
Cluster Tree: Quad-Tree on the 2D Grid

- Recursively split $N_s \times N_s$ grid at midpoints
- Each node owns a contiguous **permuted** index range
- Leaves hold $\leq$ `leaf_size` elements
- Depth $\leq \log_4 N$

Geometry (Chebyshev distance):

$$\mathrm{diam}(t) = \max(\Delta i_x, \Delta i_y)$$

$$\mathrm{dist}(t, s) = \max(d_x^+, d_y^+) \;\geq 0$$



Admissibility Condition & Block Partition

Block (t, s) is **admissible** (low-rank) when:

$$\min\big(\mathrm{diam}(t), \mathrm{diam}(s)\big) \;\le\; \eta\, \mathrm{dist}(t, s), \quad \eta = 2$$

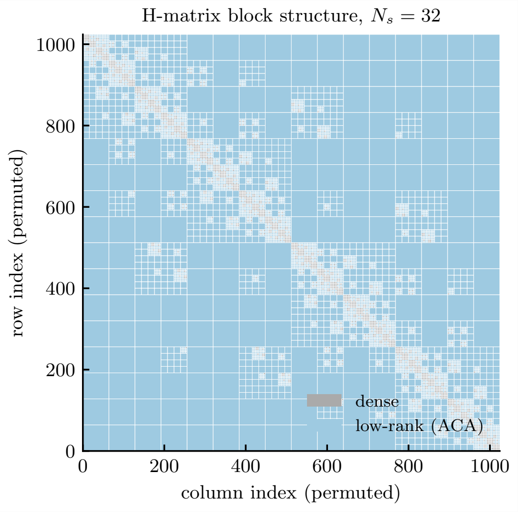
Otherwise (near-diagonal): **dense** block.

ACA fill (admissible blocks):

$$S[I, J] \approx UV^{\mathsf{T}}$$

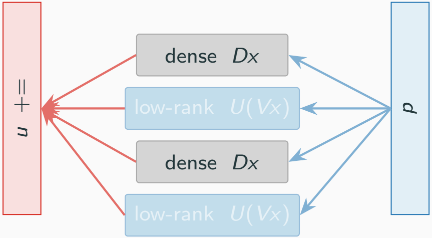
Stop when $\|u_k\|\|v_k\| \le \varepsilon_{\mathrm{aca}} \|A_k\|_F$

Typical rank $k \approx 11$ at $N_s = 64$



H-Matrix Matrix–Vector Product

Block loop (OpenMP parallel):



$$u(I) \mathrel{+}= D\,p(J) \quad \text{(dense)}$$

$$u(I) \mathrel{+}= U\big(V\,p(J)\big) \quad \text{(low-rank)}$$

- Permute p once before the loop
- Each block is **independent** $\rightarrow$ OpenMP
- Complexity: $O(N \log^2 N)$

Measured: $N = 4096$ ($N_s = 64$)

assembly: 7 ms — matvec: 0.6 ms

Boussinesq Green's Function: Love Element Formula

Green's function for elastic half-space ($E^* = E/(1-\nu^2)$):

$$G(\mathrm{r}) = \frac{1}{\pi E^* |\mathrm{r}|}$$

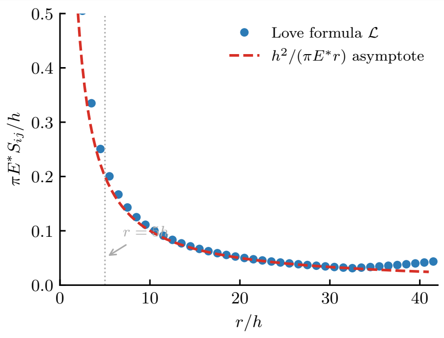
Discretise on $N_s \times N_s$ grid of elements of side h:

$$S_{ij} = \frac{1}{\pi E^*}\,\mathcal{L}\left(\Delta x, \Delta y, \tfrac{h}{2}\right)$$

Love (1929) exact integral over a square element:

$$\mathcal{L}(x,y,a) = (x+a)\ln\frac{(y+a)+R_{++}}{(y-a)+R_{+-}} + (y+a)\ln\frac{(x+a)+R_{++}}{(x-a)+R_{-+}}$$
$$+ (x-a)\ln\frac{(y-a)+R_{--}}{(y+a)+R_{-+}} + (y-a)\ln\frac{(x-a)+R_{--}}{(x+a)+R_{+-}}$$

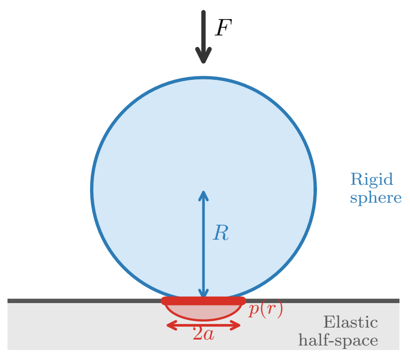
Self-term: $S_{ii} = \dfrac{4h\ln(1+\sqrt{2})}{\pi E^*}$



3.7 % deviation at $r = h$; below 0.2 % at $r \geq 5h$

The Hertz Problem

Hertz Contact: Problem Setup



Rigid sphere (radius R) pressed onto an **elastic half-space** (E^*) with total load F.

Parabolic **initial gap**:

$$g_0(r) = \frac{r^2}{2R}, \qquad r = \sqrt{x^2 + y^2}$$

Unknowns: contact pressure $p(\mathrm{x})$, contact radius a, approach δ.

Boundary conditions in contact ($r < a$)

$p \geq 0, \quad u_z = \delta - g_0 \geq 0$

Outside contact ($r \geq a$)

$p = 0, \quad u_z < \delta - g_0$

Hertz Analytical Solution

Contact radius:

$$a = \left(\frac{3FR}{4E^*}\right)^{1/3}$$

Hertzian pressure distribution:

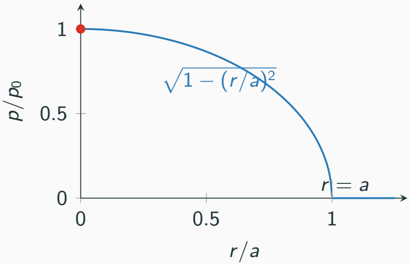
$$p(r) = p_0\sqrt{1 - \left(\frac{r}{a}\right)^2}, \quad r \le a$$

Peak pressure:

$$p_0 = \frac{3F}{2\pi a^2}$$

Indentation:

$$\delta = \frac{a^2}{R}$$



BEM Discretisation & Polonsky–Keer Algorithm

Grid $N_s \times N_s$, element size $h = L/N_s$, flat index $k = i_y N_s + i_x$:

$$S_{ij} = \frac{\mathcal{L}(\mathrm{x}_i - \mathrm{x}_j,\ h/2)}{\pi E^*}$$

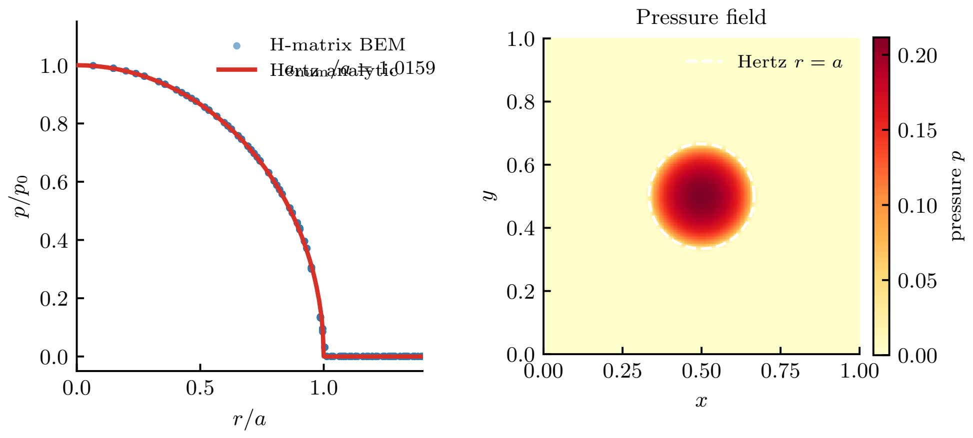
Contact QP:

$$\min_p \ \tfrac{1}{2} p^\mathsf{T} S p + p^\mathsf{T} g_0 \text{ s.t. } p_i \geq 0,\ \bar{p} = p_{\mathrm{nom}}$$

Polonsky–Keer (1999) PCG

1. Init $p = \bar{p}1$; $u = Sp$; $r = u + g_0$
2. Shift: $\bar{r} = r - \langle r \rangle_{\mathrm{contact}}$
3. Zero r_i if $p_i = 0$ and $\bar{r}_i \geq 0$
4. $\delta = \|r\|^2$; **converge** if $\delta < \varepsilon^2 \delta_0$
5. $t = r + (\delta/\delta_{\mathrm{old}})\, t$; mask t
6. $v = St$; $\tau = \langle r, t \rangle / \langle v, t \rangle_{\mathrm{c}}$
7. $p \mathrel{-}= \tau t$; $p \leftarrow \max(p, 0)$
8. Overlap correction; normalise p
9. $u = Sp$; $r = u + g_0$

Hertz Validation ($N_s = 64,\ E^* = 1,\ R = 0.5$)



$a_{\mathrm{num}}/a_{\mathrm{Hertz}} = 1.016$ $\quad p_{\max}/p_0 = 0.998$

Mean pressure error $< 10^{-10}$ $\quad$ Converged in 29 iterations

Rough Surface Contact

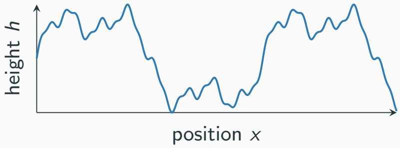
Real Surfaces are Multi-Scale Rough

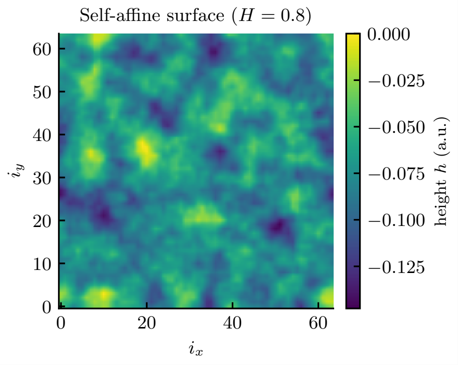
Self-affine surfaces: statistically scale-invariant

$$C(q) \sim q^{-2(1+H)}, \quad H \in (0,1)$$

(H = Hurst exponent, q = spatial frequency)

Profile sketch (three-scale superposition):





$N_s = 64$, $H = 0.8$, seed 12345

normalised: RMS slope = 1

Normal Contact as a Quadratic Programme

Influence matrix: $u = Sp$, symmetric PSD $(N \times N)$

Complementarity energy:

$$W(p) = \frac{1}{2} p^{\mathsf{T}} S p + p^{\mathsf{T}} g_0$$

Minimise subject to:

$$p_i \geq 0 \quad \forall i, \qquad \frac{1}{N} \sum_i p_i = \bar{p}$$

KKT conditions (complementarity):

$$(Sp + g_0)_i \geq 0, \quad p_i \geq 0$$

$$p_i \, (Sp + g_0)_i = 0$$

Physical meaning

- $p_i > 0$: element *in contact* $\Rightarrow$ gap $= 0$
- $p_i = 0$: element *not in contact* $\Rightarrow$ gap ≥ 0

No-penetration + non-adhesion

$g_i = u_i + g_{0,i} - \delta \geq 0$ everywhere, with δ the rigid-body approach.

Polonsky–Keer Projected CG (full)

Initialisation

$p \leftarrow \bar{p}1; \quad u \leftarrow Sp; \quad r \leftarrow u + g_0$

Iteration $(k = 1, \ldots)$

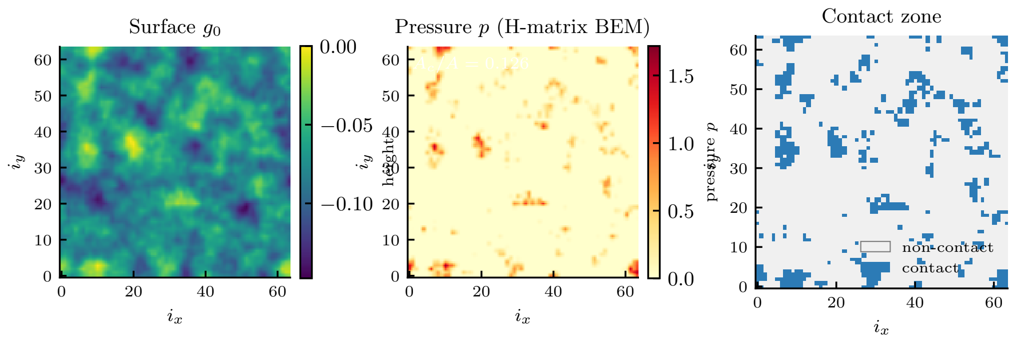
1. Shift gradient: $\bar{r} \leftarrow r - \langle r \rangle_{\mathcal{C}}$
2. Suppress: $r_i \leftarrow 0$ if $p_i = 0$ and $\bar{r}_i \geq 0$
3. $\delta \leftarrow \|r\|^2$; **converge** if $\delta < \varepsilon^2 \delta_0$
4. CG direction: $t \leftarrow r + \frac{\delta}{\delta_{\mathrm{prev}}} t$ (masked)
5. Line search: $\tau = \delta \,/\, \langle t, St \rangle_{\mathcal{C}}$
6. Update: $p \leftarrow \max(p - \tau t, 0)$
7. Overlap: $p_i \mathrel{-}= \tau g_i$ if $p_i = 0,\ g_i < 0$
8. Normalise: $p \leftarrow p \cdot \bar{p} N / \sum p$
9. Recompute $u = Sp$; $r = u + g_0$

Convergence criterion

$$E_k = \frac{\sum_i p_i |g_i|}{N \,\bar{p}\, \Delta g} < \varepsilon$$

- **Two H-matvecs per step**: one for gradient update, one for the line search
- Overlap correction resets CG $(\delta/\delta_{\mathrm{prev}} \leftarrow 0)$
- Converges in $\lesssim 30$ iterations for typical rough surfaces at $\bar{p} = 0.05\, E^*$

Rough Surface Contact Result ($N_s = 64$, $\bar{p} = 0.05$)



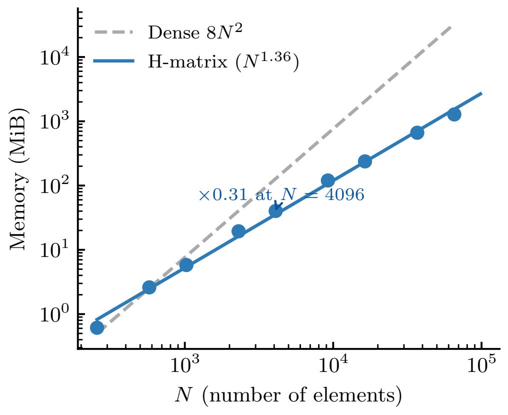
Self-affine surface ($H = 0.8$)

$A_c/A = 0.126$, mean $p = 0.05\,E^*$

Contact in 12.6 % of elements

Performance & Comparison

Memory Scaling: Dense vs. H-Matrix



Key numbers at $N = 4096$ ($N_s = 64$)

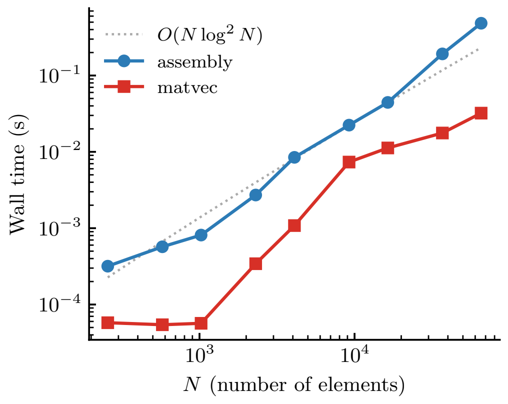
Quantity	Value
Dense storage	128 MiB
H-matrix storage	40 MiB
Compression	**0.31**$\times$
Dense blocks	2 116
Low-rank blocks	6 900
Avg. rank k	11

At $N_s = 256$ ($N = 65\,536$):

H-matrix: 1.3 GB vs. Dense: 34 GB

Fit exponent $\approx N^{1.45}$ in practice (target $O(N \log^2 N) \approx N^{1.28}$ at $N = 4096$)

Timing Scaling: Assembly & Matvec vs. N

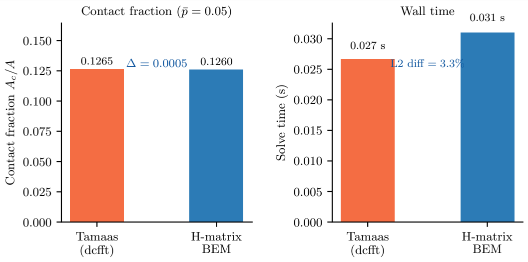


Measured wall times (20-core Xeon)

N_s	Assembly	Matvec
64	7 ms	0.6 ms
128	44 ms	11 ms
256	484 ms	32 ms

- Assembly parallelised with OpenMP
- Matvec: 9 016 block GEMVs (independent $\to$ OpenMP)
- Both roughly $O(N \log^2 N)$
- Matvec dominated by low-rank blocks ($k \approx 11$)

Comparison with Tamaas (Non-Periodic)



Summary ($N_s = 64$, $\bar{p} = 0.05$)

Metric	Tamaas	BEM
A_c/A	0.1265	0.1260
$\lvert\Delta A_c/A\rvert$	0.0005	
$\langle p\rangle/\bar{p} - 1$	$< 10^{-14}$	$< 10^{-14}$
Iters	33	26
$\lVert p_{\mathrm{BEM}} - p_{\mathrm{T}}\rVert / \lVert p_{\mathrm{T}}\rVert$	3.3 %	

The 3.3 % L2 difference is dominated by Gibbs-like near-field errors in Tamaas dcfft operator ($\pm$2–8 % at $r = 1$–$3h$), *not* by the H-matrix approximation.

Conclusions

Delivered:

- C++17 H-matrix BEM library (`libhmatrix_contact + Python module`)
- Love element kernel (exact, $O(1)$ lookup)
- Quad-tree + ACA ($\varepsilon_{\mathrm{aca}} = 10^{-6}$)
- Polonsky–Keer PCG solver
- OpenMP-parallel assembly & matvec

Validated:

- Hertz: $a_{\mathrm{num}}/a = 1.016$, $p_{\mathrm{max}}/p_0 = 0.998$
- H-matvec vs. dense: $< 10^{-5}$ rel. L2
- Rough contact vs. Tamaas: $|\Delta A_c| = 0.0005$

Outlook

- Larger grids via MPI ($N_s \sim 10^3$, $N \sim 10^6$)
- Plasticity (Mindlin–Deresiewicz)
- Adhesion (JKR / DMT)
- 3D bodies (volume BIE)
- Non-conforming surface meshes

Code: `Hcontact/`

Build:

- `cmake -S . -B build -DCMAKE_CXX_COMPILER=/usr/bin/g++`
- `cmake --build build -j`
- `ctest --test-dir build`